西北农林科技大学

Northwest A&F University

流式计算实验报告

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环境准备

```
root@master:~# /etc/init.d/mysql start
 * Starting MySQL database server mysqld
su: warning: cannot change directory to /nonexistent: No such file or directory
                                                                        [ OK ]
root@master:~# start-cluster.sh
Starting cluster.
Starting standalonesession daemon on host master.
Please specify JAVA_HOME. Either in Flink config ./conf/flink-conf.yaml or as sy
stem-wide JAVA_HOME.
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stem-wide JAVA_HOME.
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stem-wide JAVA_HOME.
root@master:~#
```
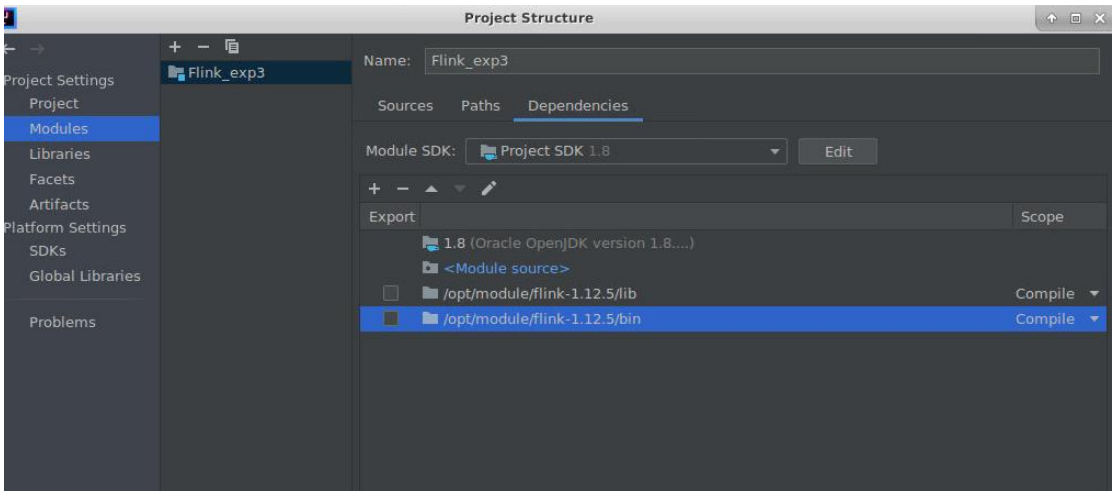

导入依赖



基于时间的滚动和滑动窗口

设置并行度

```
env.setParallelism(2);
```

实验结果

输入

```
root@cg:~# nc -l -k -p 9999
9,3
9,2
9,7
4,9
2,6
1,5
2,3
5,7
5,4
```

输出 1

```
res1:2> WindowDemo01_TimeWindow.CartInfo(sensorId=9, count=5)
res1:1> WindowDemo01_TimeWindow.CartInfo(sensorId=4, count=9)
res1:2> WindowDemo01_TimeWindow.CartInfo(sensorId=9, count=7)
res1:1> WindowDemo01_TimeWindow.CartInfo(sensorId=2, count=6)
res1:1> WindowDemo01_TimeWindow.CartInfo(sensorId=1, count=5)
res1:1> WindowDemo01_TimeWindow.CartInfo(sensorId=2, count=3)
res1:2> WindowDemo01_TimeWindow.CartInfo(sensorId=5, count=11)
```

输出 2

```
res2:2> WindowDemo01_TimeWindow.CartInfo(sensorId=9, count=3)
res2:2> WindowDemo01_TimeWindow.CartInfo(sensorId=9, count=12)
res2:2> WindowDemo01_TimeWindow.CartInfo(sensorId=9, count=9)
res2:1> WindowDemo01_TimeWindow.CartInfo(sensorId=4, count=9)
res2:1> WindowDemo01_TimeWindow.CartInfo(sensorId=4, count=9)
res2:1> WindowDemo01_TimeWindow.CartInfo(sensorId=1, count=5)
res2:1> WindowDemo01_TimeWindow.CartInfo(sensorId=2, count=9)
res2:2> WindowDemo01_TimeWindow.CartInfo(sensorId=5, count=11)
res2:1> WindowDemo01_TimeWindow.CartInfo(sensorId=1, count=5)
res2:1> WindowDemo01_TimeWindow.CartInfo(sensorId=2, count=9)
INFO  [] - Shutting down BLOB cache
```

可以发现，滑动窗口由于窗口之间重叠，导致了数据统计也有重叠，而滚动窗口则无

基于数量的滚动和滑动窗口

设置并行度

```
env.setParallelism(2);
```

实验结果

输入

```
root@cg:~# nc -l -k -p 9999
9,3
9,2
9,7
4,9
2,6
1,5
2,3
5,7
5,4
2,1
9,3
9,2
```

输出

```
2> WindowDemo02_TimeWindow.CartInfo(sensorId=9, count=12)
1> WindowDemo02_TimeWindow.CartInfo(sensorId=4, count=9)
1> WindowDemo02_TimeWindow.CartInfo(sensorId=1, count=5)
1> WindowDemo02_TimeWindow.CartInfo(sensorId=2, count=9)
2> WindowDemo02_TimeWindow.CartInfo(sensorId=5, count=11)
1> WindowDemo02_TimeWindow.CartInfo(sensorId=2, count=1)
2> WindowDemo02_TimeWindow.CartInfo(sensorId=9, count=5)
```

会话窗口

设置并行度

```
env.setParallelism(2);
```

实验结果

输入

```
root@cg:~# nc -l -k -p 9999
1,3
1,5
2,4
3,4
```

输出

```
1> WindowDemo03_TimeWindow.CartInfo(sensorId=1, count=8)
1> WindowDemo03_TimeWindow.CartInfo(sensorId=2, count=4)
1> WindowDemo03_TimeWindow.CartInfo(sensorId=3, count=4)
```